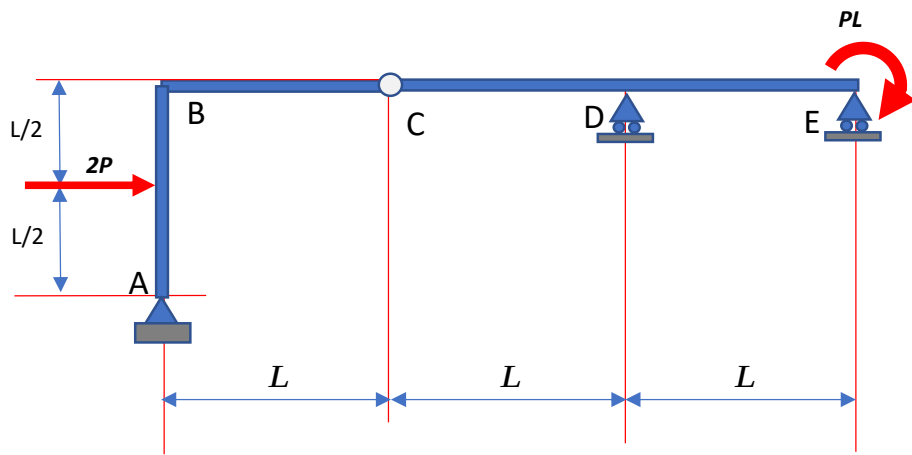


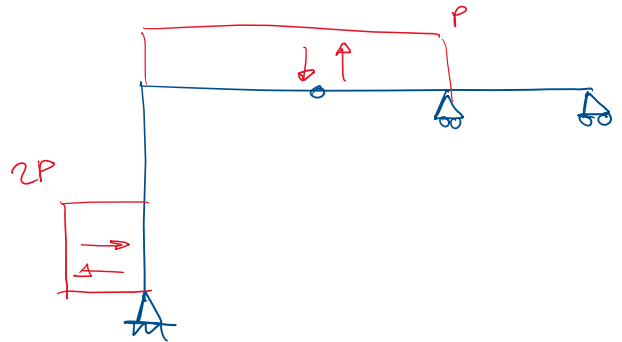
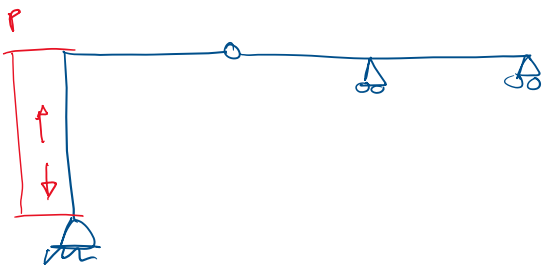
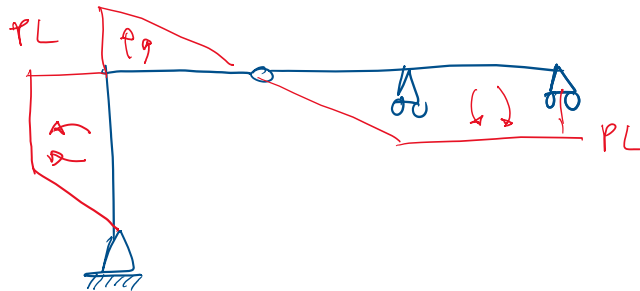


a) Reacciones :

$$R_{yE} = 0 \quad R_{yA} = -P$$

$$R_{yD} = P \quad R_{xA} = -2P$$

b) Diagramas :



c) MOVIMIENTO $\vec{u}_F$

$$\uparrow V_E = \uparrow V_D + \vec{\Theta}_D \cdot L - \frac{PL \cdot L}{EI} \cdot \frac{L}{2} \Rightarrow \vec{\Theta}_D = \frac{PL^2}{2EI}$$

$$\uparrow V_C = \uparrow V_D + \vec{\Theta}_D \cdot L - \frac{PL \cdot L}{2EI} \cdot \frac{2}{3}L = -\frac{PL^2}{2EI}L - \frac{PL^3}{3EI} = -\frac{5}{6} \frac{PL^3}{EI}$$

$$\uparrow V_c = \cancel{\uparrow V_A} + \hat{\Theta}_A \cdot L + \frac{PL \cdot L/2}{2EI} L + \frac{PL \cdot L/2}{EI} L + \frac{PL \cdot L}{2EI} \frac{2}{3} L = -\frac{5 PL^3}{6 EI}$$

$$\hat{\Theta}_A = \frac{PL^2}{4EI} + \frac{PL^2}{2EI} + \frac{PL^2}{3EI} + \frac{PL^2}{EI} \frac{5}{6} = \frac{3+6+4+10}{12} \frac{PL^2}{EI} = \frac{23 PL^2}{12 EI}$$

$$\vec{U}_F = \cancel{\vec{U}_A} + \hat{\Theta}_A \frac{L}{2} - \frac{PL \cdot L/2}{2EI} \cdot \frac{1}{3} \frac{L}{2} = \frac{23 PL^2}{12 EI} \frac{L}{2} - \frac{PL^3}{24} = \frac{11 PL^3}{12 EI}$$

SEGUNDA PARTE

Reacciones y diagramas son iguales $\rightarrow$ EST. ISOSTÁTICA

Separo $U_E \rightarrow U_E^{MECAN} + U_E^{TERMICO}$

$$\vec{U}_E^{MEC} = \vec{U}_B = \hat{\Theta}_A \cdot L - \frac{PL \cdot L/2}{2EI} \left(\frac{L}{2} + \frac{1}{3} \frac{L}{2} \right) - \frac{PL \cdot L/2}{EI} \frac{L}{4} =$$

$$\vec{U}_E^{MEC} = \frac{23 PL^2}{12 EI} L - \frac{PL^3}{6EI} - \frac{PL^3}{8EI} = \frac{PL^3}{EI} \frac{46-4-3}{24} = \frac{39 PL^3}{24 EI}$$

$U_E^{TERMICO} \rightarrow$ Al no haber esfuerzos $V_c^{TERMICO} = \emptyset$

$$\uparrow V_B^{TEN} = \alpha L \Delta T$$

$$\hat{\Theta}_A^{TEN} = \hat{\Theta}_B^{TEN} = \hat{\Theta}_C^{TEN}$$

$$\uparrow V_B^{TEN} = \hat{\Theta}_C \cdot L \Rightarrow \hat{\Theta}_C^{TEN} = \alpha \Delta T$$

$$\vec{U}_B^{TEN} = \vec{U}_E^{TEN} = \hat{\Theta}_A \cdot L = \alpha \Delta T L = U_B^{TEN}$$

$$\vec{U}_E^{TOT} = \vec{U}_E^{MEC} + U_E^{TEN} = \frac{39 PL^3}{24 EI} + \alpha \Delta T L$$

